NYU TANDON SCHOOL OF ENGINEERING

ESSENTIALS OF CLOUD SECURITY - FUNDAMENTALS, TECHNIQUES AND APPLICATIONS

OVERVIEW

The rapid shift to cloud computing in recent years has enabled incredible advances and convenience, but it also comes with new implications for risk and the need for increased security—from user access, authentication measures, and key management to data security, legal issues, and policies associated with virtual environments.

This program is designed to teach participants how cloud computing has evolved into a standard technology model across industries and how to use cloud security tools. Through hands-on lab activities, you will build a virtual cloud environment and learn key cloud security concepts, including storage methods and controls, risk management techniques, access and authentication management, encryption and key management, automation and alert systems, and zero trust policies. What's more, you'll get real-world experience using cloud security tools and have the ability to showcase your work as you build skills you can apply in your own organization.

Through this program, you will:

- Master the essentials of cloud security via a curriculum based on the NIST framework, and learn to identify foundational technologies used in cloud computing and security
- Experience hands-on learning via Amazon Web Services (AWS) cloud computing service which is extensively used by organizations, and apply access management practices to identity, authentication, and authorization management
- Learn to address key issues pertaining to data breaches with zero trust principles
- Earn a certificate from NYU Tandon Online

Program Duration
6 Weeks, Online | 4–6 hours per week

Program Fee
US$1,500

DEFENDING THE CLOUD

Cloud computing is on the rise—and so is the need for skilled professionals to manage and secure it.

$1,712.44 billion

In global spending on cloud services is predicted by 2029, up from $313 billion in 2020, with a CAGR of 19.9%.

SOURCE: Fortune Business Insights

115%

Predicted a growth in demand for cloud security skills over the next five years.

Source: Burning Glass [AT3]

56%

Of organizations struggle to find employees with the skills needed to deploy and manage consistent security across multi-cloud environments.

Source: Check Point's 2022 Cloud Security Report[AT4]

50%

Of all organizations have difficult understanding how security solutions work together in multi-cloud environments that involve disparate solutions from multiple vendors.

SOURCE: Check Point's 2022 Cloud Security Report

1.3 million

Professionals are in roles pertaining to network security, cybersecurity, penetration testing, DevOps, applications engineering, and IT specialists, yet just 8,000 list cloud security as a skill set.

SOURCE: LinkedIn

79%

Of organizations report experiencing at least one cloud data breach in the last 18 months, and 43% report experiencing 10 or more breaches in that time (since March 2022).

Source: Ermetic

KEY TAKEAWAYS

In a Security Magazine survey of IT leaders, 95% of respondents reported that multi-cloud was a strategic priority for their organizations in 2022. The pandemic helped to fuel this growth, as more organizations moved to a hybrid/remote work model and are continuing to lean on the cloud for secure data storage.

Throughout the program, you will:

- **Explore** the foundational concepts of cloud computing, including the history and development of the cloud platform
- **Identify** foundational technologies used in cloud computing and security
- **Assess** policies and control systems that govern cloud services
- **Learn** to apply access management practices to identity, authentication, and authorization management
- **Apply** concepts of cloud computing security to a virtual cloud environment
- **Gain** experience using cloud computing security tools, and develop the skills to apply them in your own organization

WHO IS THIS PROGRAM FOR?

This program is designed for individuals looking to take a deep dive into cloud security and understand the safety protocols that are critical to cloud computing. It is ideal for:

- **Engineers and IT Professionals** seeking to understand the foundational concepts of cloud computing and the types of technologies used in cloud computing and security and align their efforts with business requirements and priorities

- **Early to mid-level cybersecurity professionals** looking to apply concepts of cloud computing security to a virtual cloud environment and facilitate the smooth migration of their organization's systems to the cloud with the necessary security measures and risk management plans

- **IT/cybersecurity consultants** aiming to offer strategic and feasible long-term IT-based solutions and ensure their teams are working optimally and keeping their skills up to date

PROGRAM SYLLABUS

The essentials of cloud security - fundamentals, techniques and applications program provides a true hands-on experience, with hands-on lab activities built into each module that enable you to create a virtual cloud environment by applying the concepts presented while also showcasing your work on the cloud.

Module 1: Cloud Foundation Concepts

- Learn about the foundational technology and concepts that make cloud computing possible
- Identify the history and fundamentals of cloud computing environments
- Determine best use of virtualization models
- Understand the transition from server to serverless
- Create a container in a cloud computing environment

Hands-On Lab Assignment: Create a Lambda function in Amazon Web Services (AWS), launch an Elastic Compute Cloud (EC2) instance, and create a Simple Storage Service (S3) storage bucket.

Module 2: Cloud Data Storage Controls and Risk Management Basics

- Identify data technologies in cloud environments
- Describe security policies and data security technologies
- Assess risk in a cloud environment
- Apply security measures to a cloud environment

Hands-On Lab Assignment: Create encryption and key management measures in cloud environment using Vault.

Module 3: Identity and Access Management in the Cloud

- Discuss attributes of access management technologies
- Compare various authentication and authorization technologies
- Secure credentials in a cloud environment

Hands-On Lab Assignment: Create authentication measures using Guard Duty.

Module 4: Cloud Crypto and Key Management

- Review features of modern cryptography
- Compare the various forms of cryptography used in cloud computing
- Discuss message integrity and digital signatures
- Examine encrypting data and using keys in the public cloud
- Apply key management systems in a cloud environment

Hands-On Lab Assignment: Learn how to use Amazon's Elastic Kubernetes Service (EKS) and automate the deployment, scaling, and management of containerized applications within your AWS infrastructure.

Module 5: Automation and DevSecOps

- Define principles and uses of DevOps
- Discuss the problems with the traditional security model that led to the evolution of DevSecOps
- Create an automated response in the cloud pipeline for compromised data

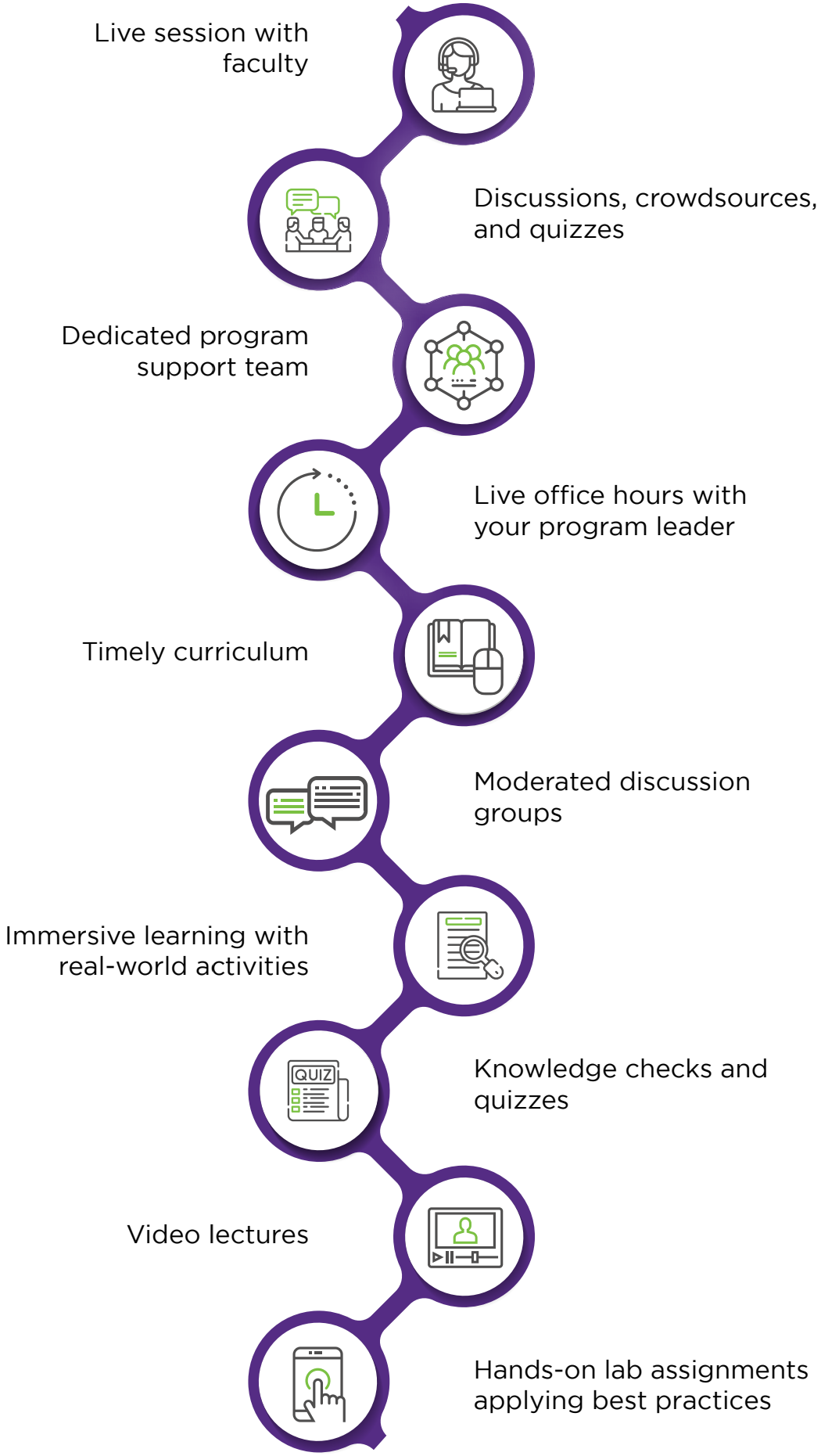
Hands-On Lab Assignment: Create an alert in Amazon Web Services (AWS).

Module 6: Zero Trust

- Examine zero trust principles
- Evaluate the tenets of zero trust networks
- Discuss the various zero trust architecture models
- Showcase a simulated cloud environment that uses the tools presented in the program

Final Project: Create a short presentation showcasing the completed labs.

THE LEARNING EXPERIENCE



PROGRAM FACULTY

Dean Christakos

Adjunct Assistant Professor and Computer Scientist

Dean Christakos has extensive research experience in fields from wearable computing and embedded systems to security architecture and big data analytics. With research work in securing mobile and wireless networks, he joined the Johns Hopkins University Applied Physics Laboratory, where he worked with clients at the Department of Defense to build more secure architectures for their networks and analytics results. Following this, he served as Vice President in the Surveillance Analytics group within Technology Risk at Goldman Sachs, a group that demonstrated how to use big data from activity around the organization to find patterns indicating security violations. Professor Christakos has an MS and PhD from the MIT Media Laboratory, an MS from Carnegie Mellon's Department of Electrical and Computer Engineering, and a BS in Computer Science from MIT.

CERTIFICATE

Upon successful completion of the program, you will be awarded a verified digital certificate of participation from NYU Tandon.

ABOUT

NYU TANDON SCHOOL OF ENGINEERING

The NYU Tandon School of Engineering empowers people to use science and technology as tools to build a better society. Dedicated to creating solutions that can tackle tomorrow's problems today, the school focuses on invention, innovation, and entrepreneurship.

As an institution that traces its origins back to 1854, NYU Tandon School of Engineering is rooted in tradition and opportunity. The school was, quite literally, the birthplace of the American Dream; it was its alumnus, James Truslow Adams, who coined that phrase in 1931. Just as it was then, NYU Tandon School of Engineering is a place where ambitious students from all walks of life get the solid education they need to launch their careers.

Invention, innovation, and entrepreneurship suffuse every aspect of NYU Tandon as it strives to carry the school's great history forward with a shared goal: to generate ideas and build solutions for a healthier, safer, better-connected, and more sustainable world.

ABOUT EMERITUS

New York University is collaborating with online education provider Emeritus to offer executive education programs. An Emeritus Certificate program created in collaboration with New York University is based on syllabus created by New York University, and contains video lecture content created and recorded by New York University faculty, combined with assessments, assignments, projects, cases, and exercises created in collaboration with faculty and subject matter experts, and delivered by Emeritus.

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